Final Project

Critical Evaluation and Empirical Stress Test
of a Causal Inference Paper

Causal Inference in the AI Era

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Paper Information

RATE: Causal Explainability of Reward Models with Imperfect Counterfactuals

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1 Summary of the Paper

Reward models (RMs) play a central role in the alignment and evaluation of large language models. They are used to rank candidate responses, guide post-training, support best-of-n selection, and evaluate model outputs. Despite their importance, reward models are typically treated as black boxes: a model assigns a numerical score to a prompt-response pair, but this score does not directly reveal which properties of the response influenced the evaluation.

The paper studies whether high-level textual attributes, such as helpfulness, sentiment, complexity, or length, have a causal effect on the score assigned by a reward model. A simple comparison between responses with and without an attribute only measures association. Such a comparison may be confounded by other response characteristics, including grammar, style, length, or formatting. Consequently, it may suggest that a reward model is sensitive to the target attribute even when the observed difference is actually driven by correlated off-target properties.

To address this problem, the authors formulate reward-model explainability as a causal inference task. The target quantity is the change in reward that would occur if one response attribute were modified while all other aspects of the response remained fixed. Since both potential versions of the same response cannot normally be observed, the paper uses an LLM to rewrite responses and generate approximate counterfactuals. However, these rewrites are imperfect because the LLM may change off-target attributes in addition to the intended treatment.

The paper proposes the *Rewrite-based Attribute Treatment Estimator* (RATE), which applies a double-rewrite procedure. A response is first rewritten to reverse the target attribute and then rewritten again toward its original attribute value. RATE compares the first rewrite with the rewrite-of-the-rewrite, rather than directly comparing the original response with a single rewrite. Under assumptions concerning the distribution of rewrite errors and the structure of the reward function, the effects of off-target changes cancel in expectation, allowing RATE to estimate the causal effect of the target attribute.

The authors provide theoretical identification results and evaluate RATE in both semi-synthetic and real reward-model settings. Their experiments show that naive observational comparisons and single-rewrite estimators can be strongly influenced by confounding or imperfect rewrites, whereas RATE can recover the intended causal effect when its assumptions hold. The paper therefore contributes a causal framework for reward-model explainability, together with an estimator designed specifically for settings in which only imperfect LLM-generated counterfactuals are available.

2 Causal Setup

The unit of analysis is a prompt-response pair (X_i, Y_i), where X_i is the prompt and Y_i is the corresponding response. Let $W_i \in \{0, 1\}$ denote whether the response possesses the target textual attribute, such as positive sentiment, helpfulness, or high complexity. The outcome is the reward-model score $R(X_i, Y_i)$ assigned to that prompt-response pair.

Following the potential-outcomes framework, $Y_i(1)$ and $Y_i(0)$ denote the two potential versions of the response under the presence and absence of the target attribute. These responses should differ only in W, while all other relevant properties remain fixed. The corresponding potential outcomes are the reward scores $R(X_i, Y_i(1))$ and $R(X_i, Y_i(0))$.

The primary estimand is the Average Treatment Effect (ATE):

$$\text{ATE} = \mathbb{E}\left[R(X, Y(1)) - R(X, Y(0))\right]. \tag{1}$$

The ATE represents the expected change in the reward-model score if the target attribute were changed from $W = 0$ to $W = 1$ while all other properties of the response remained unchanged. The paper also considers the Average Treatment Effect on the Treated (ATT) and the Average Treatment Effect on the Untreated (ATU):

$$\text{ATT} = \mathbb{E}\left[R(X, Y(1)) - R(X, Y(0)) \mid W = 1\right], \tag{2}$$

$$\text{ATU} = \mathbb{E}\left[R(X, Y(1)) - R(X, Y(0)) \mid W = 0\right]. \tag{3}$$

These quantities may differ because a reward model can respond differently to changing an attribute among responses that originally possess it and responses that do not. RATE estimates the ATT and ATU separately and combines them according to the proportions of treated and untreated observations:

$$\text{ATE} = P(W = 1)\text{ATT} + P(W = 0)\text{ATU}. \tag{4}$$

For each observed response, only one potential response, $Y_i(W_i)$, is available. The alternative response is an unobserved counterfactual. RATE uses an LLM-based rewrite operation, denoted by $\text{Re}(X, Y, w)$, to approximate this missing response by rewriting Y toward the desired attribute value w.

The paper further distinguishes between off-target attributes Z, which remain unchanged during rewriting, and rewrite-sensitive attributes ξ, which may be unintentionally modified. Examples of such changes include grammar, length, tone, or formatting. These rewrite-induced changes are important because they may also influence the reward score and therefore bias a causal estimate based on rewritten responses.

3 Key Assumptions

RATE represents a response as $Y(W, Z, \xi)$, where W is the target attribute, Z contains off-target attributes that remain unchanged during rewriting, and ξ contains off-target attributes that may be unintentionally modified.

3.1 Assumption 1: Direction-Independent Rewrite Errors

The off-target changes introduced by rewriting are assumed to follow a distribution that does not depend on the rewrite direction:

$$\text{Re}(X, Y(W, Z, \xi), 1 - W) \overset{d}{=} Y(1 - W, Z, \widetilde{\xi}), \qquad \widetilde{\xi} \sim P_{\text{Re}}. \tag{5}$$

Thus, rewrites from $W = 0$ to $W = 1$ and from $W = 1$ to $W = 0$ may introduce different errors in individual examples, but these errors must follow the same distribution. This symmetry allows the expected effect of rewrite errors to cancel in the double-rewrite comparison.

3.2 Assumption 2: Additivity of the Reward

The reward is assumed to decompose into separate contributions from the target and immutable attributes, and from the rewrite-sensitive attributes:

$$R(X, Y(W, Z, \xi)) = R_{W,Z}(X, W, Z) + R_\xi(X, \xi). \tag{6}$$

This assumption excludes interactions between rewrite-induced changes ξ and the target or immutable attributes (W, Z). Therefore, the reward contribution of the rewrite errors can cancel independently of the causal effect of W.

3.3 Theorem 4.1: Validity of RATE

If the reward function is bounded, Assumptions 1 and 2 hold, the observations are sampled i.i.d., and $0 < P(W = 1) < 1$, then RATE provides unbiased and $\sqrt{n}$-consistent estimators of the ATT, ATU, and ATE.

The theorem establishes RATE's validity when the rewrite-error distribution is direction-independent and its reward contribution is additive. These are sufficient conditions for identification and form the theoretical basis of the double-rewrite procedure.

4 Method

The paper compares three approaches for estimating the causal effect of a textual attribute: the naive estimator, the single-rewrite estimator, and RATE.

The **naive estimator** compares the average reward of observed responses with $W = 1$ to the average reward of responses with $W = 0$:

$$\widehat{\tau}_{\text{naive}} = \frac{1}{n_1} \sum_{i:W_i=1} R(X_i, Y_i) - \frac{1}{n_0} \sum_{i:W_i=0} R(X_i, Y_i). \tag{7}$$

Because the two groups may also differ in off-target attributes, this estimator measures association rather than the isolated causal effect of W.

The **single-rewrite estimator** uses an LLM to rewrite each response toward the opposite treatment value. It compares the original response with its rewrite, approximating the missing counterfactual. However, this comparison may remain biased because the rewrite can modify off-target attributes in addition to W.

RATE addresses this problem through a **double-rewrite procedure**:

$$Y_i \longrightarrow \mathrm{Re}(Y_i, 1 - W_i) \longrightarrow \mathrm{Re}(\mathrm{Re}(Y_i, 1 - W_i), W_i)\,. \tag{8}$$

Instead of comparing the original response with the first rewrite, RATE compares the first rewrite with the rewrite-of-the-rewrite. Both responses have passed through the rewriting process, so under Assumptions 1 and 2, the expected contribution of off-target rewrite errors cancels.

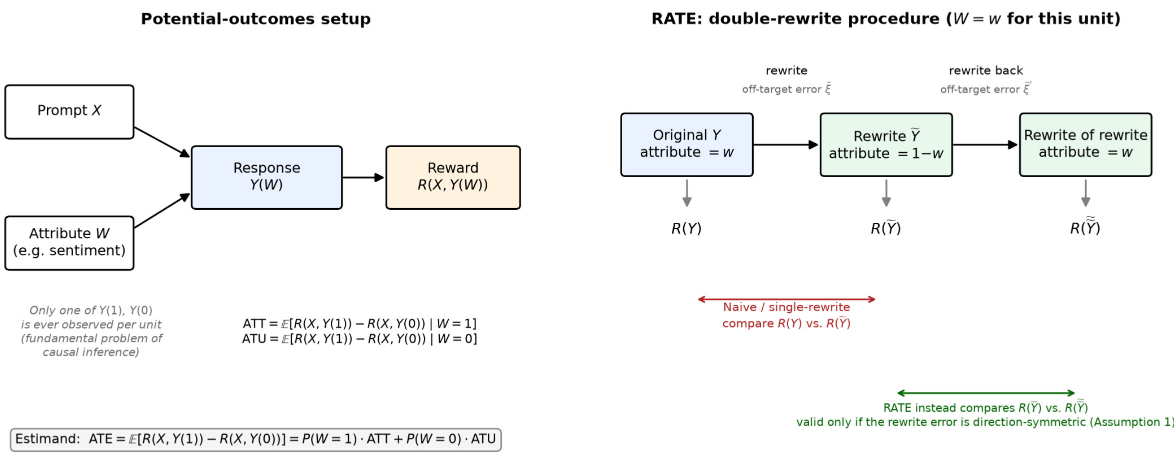

Figure 1: The potential-outcomes setup and RATE's double-rewrite procedure. The original response provides only one observed potential outcome. A first rewrite changes the target attribute, and a second rewrite returns it toward its original value. RATE compares the two rewritten responses rather than comparing an original response with a single imperfect rewrite.

For responses originally observed with $W = 1$, RATE estimates the ATT as

$$\widehat{\mathrm{ATT}}_{\mathrm{RATE}} = \frac{1}{n_1} \sum_{i:W_i=1} \left[R(X_i, \mathrm{Re}(\mathrm{Re}(Y_i, 0), 1)) - R(X_i, \mathrm{Re}(Y_i, 0)) \right]. \tag{9}$$

For responses originally observed with $W = 0$, it estimates the ATU analogously:

$$\widehat{\mathrm{ATU}}_{\mathrm{RATE}} = \frac{1}{n_0} \sum_{i:W_i=0} \left[R(X_i, \mathrm{Re}(Y_i, 1)) - R(X_i, \mathrm{Re}(\mathrm{Re}(Y_i, 1), 0)) \right]. \tag{10}$$

The final ATE estimate is the weighted combination

$$\widehat{\mathrm{ATE}}_{\mathrm{RATE}} = \frac{n_1}{n} \widehat{\mathrm{ATT}}_{\mathrm{RATE}} + \frac{n_0}{n} \widehat{\mathrm{ATU}}_{\mathrm{RATE}}. \tag{11}$$

5 Summary of the Paper's Results

The paper evaluates RATE using both semi-synthetic experiments with known or approximately known causal effects and real-world reward-model analyses.

In the first semi-synthetic experiment, typos are introduced into IMDB reviews in correlation with whether the review starts with a vowel. Since the reward model should respond to typos but not to the first letter of the review, the causal effect of “starts with a vowel” should remain close to zero. As the correlation becomes stronger, the naive and single-rewrite estimates drift away from zero, whereas RATE remains stable near the ground truth.

The second experiment examines the effect of review length on a sentiment classifier while artificially changing the correlation between length and positive sentiment. The naive estimate changes substantially under this distribution shift. RATE remains close to zero and stable, while the single-rewrite estimator develops a smaller negative bias.

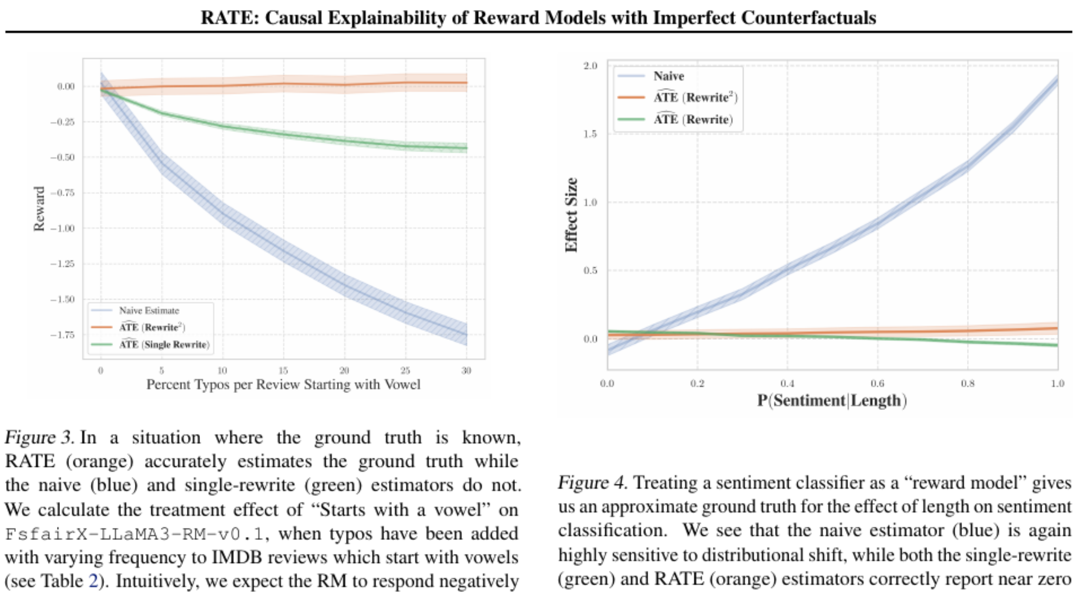

Figure 2: Semi-synthetic results from Figures 3–4 of the RATE paper. RATE remains stable as the induced correlations increase, whereas the baseline estimators are affected by confounding or imperfect rewriting.

The authors also apply RATE to several real reward models and textual attributes, including complexity, helpfulness, length, and sentiment. The RATE and naive estimates differ substantially across models and attributes. In general, the naive estimator reports larger effects for complexity, helpfulness, and length, but smaller effects for sentiment. These differences show that correlational comparisons can change the apparent interpretation of what a reward model rewards. For example, RATE suggests that the reduction in length bias reported for NCSOFT is smaller than the naive estimate indicates and may partly reflect sensitivity to correlated attributes such as complexity.

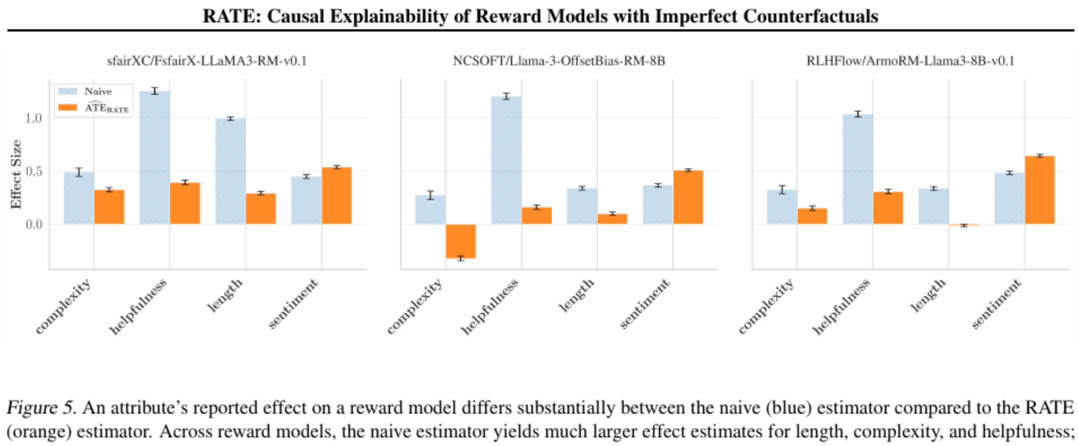

Figure 3: Naive and RATE effect estimates across three reward models and four textual attributes. The substantial differences between the estimates demonstrate that confounding can change the interpretation of reward-model behavior. Reproduced from Figure 5 of the RATE paper.

Overall, the experiments show that RATE can recover stable causal-effect estimates when its assumptions hold and that causal and correlational analyses may lead to meaningfully different conclusions about reward-model behavior.

6 Critique

RATE provides a valuable causal framework and its identification result is valid when the stated assumptions hold. However, its central limitation is the direction-independence assumption: off-target changes must follow the same distribution when rewriting from $W = 0$ to $W = 1$ and from $W = 1$ to $W = 0$.

This assumption may be reasonable for simple effects such as automatic typo correction, but it is less plausible for complex attributes. For example, making a response more helpful may require adding explanations or examples, whereas making it less helpful may require removing information. These two rewrite directions can therefore produce systematically different off-target changes.

The paper demonstrates RATE's robustness to confounding while maintaining rewrite symmetry, but it does not test how RATE behaves when this symmetry is violated. Moreover, its rewrite-quality diagnostic compares pooled reward distributions and may fail to detect differences that occur specifically between rewrite directions. Consequently, RATE does not eliminate the need for an untestable assumption; it replaces an assumption about confounding in the observed data with an assumption about the behavior of the LLM rewriter. Our empirical experiment directly stress-tests this limitation.

7 Our Experiment

7.1 Data and simulation design

We conduct two experiments using a synthetic reward function with known ground-truth ATE, which allows bias, variance, and RMSE to be computed directly. The reward model is

$$R = \beta_W W + \beta_{\text{typo}}\, \text{has\_typo} + \varepsilon, \qquad \varepsilon \sim \mathcal{N}(0, 1), \tag{12}$$

with $\beta_W = 1$ and $\beta_{\text{typo}} = -0.5$. The model is additive, so Assumption 2 holds exactly. The true ATE is $\beta_W = 1$; because the treatment effect is homogeneous, ATT, ATU, and ATE coincide.

Experiment A: qualitative replication and sanity check. We use 3,000 IMDB reviews: 1,500 beginning with a vowel ($W = 1$) and 1,500 not ($W = 0$). Typos are introduced only into the $W = 1$ reviews at six levels:

$$\text{typo\_level} \in \{0,\ 0.05,\ 0.10,\ 0.20,\ 0.30,\ 0.40\}. \tag{13}$$

The typo-affected text is treated as the confounded original observation. The rewrite and round-trip rewrite are simulated as typo-free, representing a direction-symmetric rewriter; therefore, Assumption 1 holds by construction. This is a qualitative rather than exact replication of the paper's experiment because we use a synthetic reward function with a known ATE of 1. Its purpose is to verify that the baseline estimators become biased as confounding increases while RATE remains stable.

Experiment B: direction-asymmetry stress test. We use a fully synthetic simulation controlled by two independent parameters. confound_strength determines how strongly an original-observation typo flag correlates with W. The probabilities p_{01} and p_{10} determine whether a rewrite carries a typo in the $0 \to 1$ and $1 \to 0$ directions. We define rewrite asymmetry as

$$\Delta = p_{01} - p_{10}, \tag{14}$$

and vary Δ while holding $(p_{01} + p_{10})/2$ fixed. Confounding appears only in original observations, whereas Δ appears only in rewrites; the two mechanisms are orthogonal by construction.

7.2 Experimental setup and metrics

At each grid point, we evaluate the naive, single-rewrite, and RATE estimators across independent random seeds. Experiment A uses 200 seeds. Experiment B uses 300 seeds and 2,000 simulated units per seed, with confound_strength $\in \{0, 0.25, 0.50, 0.75\}$. For an estimator $\hat{\tau}$ and true effect $\tau = 1$, we report

$$\text{Bias}(\hat{\tau}) = \mathbb{E}[\hat{\tau}] - \tau, \tag{15}$$

$$\text{Variance}(\hat{\tau}) = \mathbb{E}\left[(\hat{\tau} - \mathbb{E}[\hat{\tau}])^2\right], \tag{16}$$

$$\text{RMSE}(\hat{\tau}) = \sqrt{\mathbb{E}[(\hat{\tau} - \tau)^2]}. \tag{17}$$

8 Results

8.1 Experiment A

As typo level increases from 0 to 0.40, the naive estimator's bias grows from approximately -0.0004 to -0.117 and its RMSE from 0.039 to 0.123. Single-rewrite bias grows from approximately -0.0003 to -0.059 and its RMSE from 0.025 to 0.064. RATE's bias remains approximately -0.0004, with RMSE near 0.025 at every level. Thus, Experiment A reproduces the intended pattern and supports the correctness of our implementation.

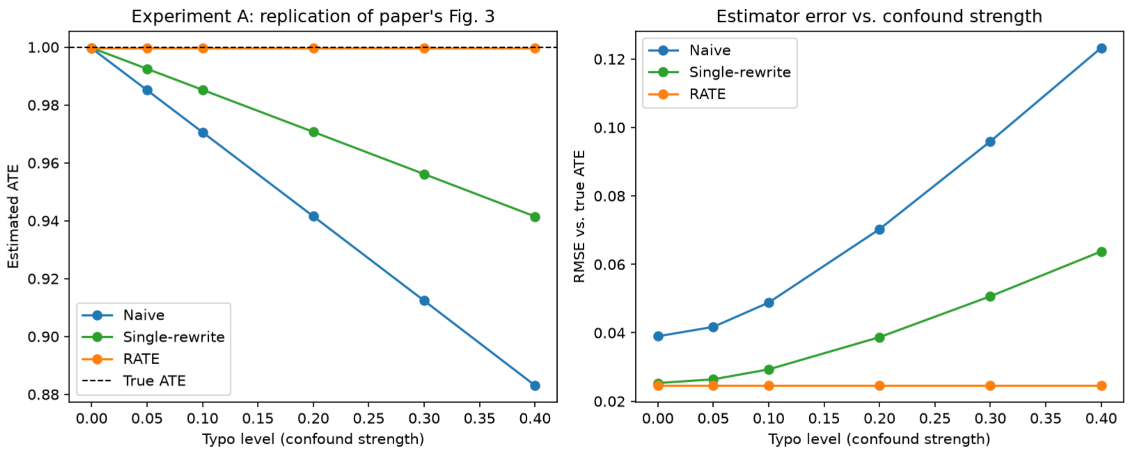

Figure 4: Estimated ATE and RMSE as typo confounding increases. The dashed line is the true ATE. RATE remains stable while naive and single-rewrite estimates degrade.

8.2 Experiment B

RATE's bias follows Δ almost linearly. The fitted slope is -0.4997, matching the closed-form prediction $\beta_{\text{typo}} = -0.5$. The relationship is unchanged across all tested confounding levels: its slope and intercept do not move as confound_strength changes. At $\Delta = 0.4$, RATE's RMSE is approximately 0.202, compared with the naive estimator's RMSE of approximately 0.045 at zero confounding and approximately 0.206 at confound_strength = 0.75. The naive estimator shows the opposite pattern: it is flat in Δ, while its bias increases in magnitude to approximately -0.20 under the strongest tested confounding. Single-rewrite is sensitive to both mechanisms.

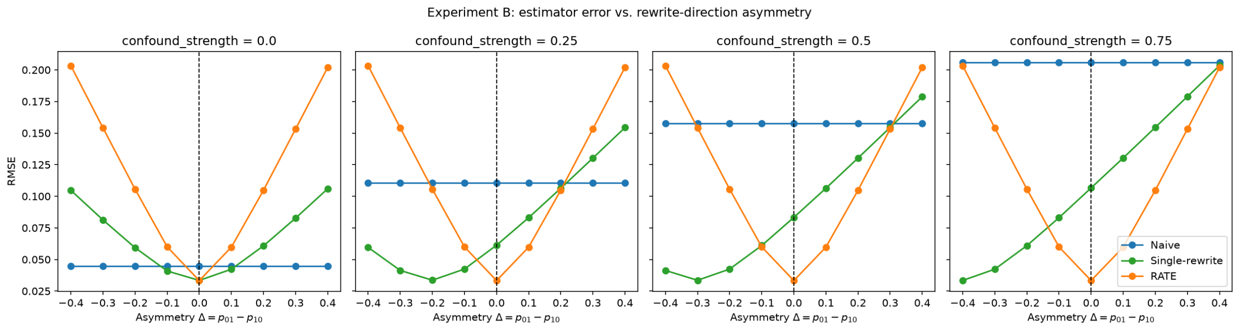

Figure 5: RMSE against the true ATE as a function of rewrite asymmetry Δ, with one panel per confounding strength. RATE is minimized at $\Delta = 0$ and is invariant to confounding; naive is invariant to Δ but worsens with confounding.

Table 1: Empirical sensitivity of the three estimators.

Estimator	Sensitive to confounding?	Sensitive to Δ?
Naive	Yes	No
Single-rewrite	Yes	Yes
RATE	No	Yes

9 Discussion

The experiment cleanly separates two vulnerabilities. RATE solves the problem for which it was designed: under $\Delta = 0$, its bias remains stable as confounding increases, as the theorem predicts. However, it is not universally more robust than the naive estimator. It exchanges sensitivity to observed-data confounding for sensitivity to direction-dependent rewrite errors. Passing the paper's confounding test provides no evidence that Assumption 1 holds, because the mechanisms are orthogonal in our design and behave orthogonally in the results.

The magnitude is practically important. At the largest asymmetry tested, $\Delta = 0.4$, RATE's error was comparable to the naive estimator's error under the strongest confounding condition. This value was selected as part of our simulation and should not be interpreted as an estimate of the asymmetry exhibited by GPT-4o or any other real rewriter. Our result therefore does not establish that such a large violation occurs in practice. Instead, it demonstrates that if direction-dependent rewrite errors occur, RATE's bias can become comparable in magnitude to the confounding bias that the method was designed to remove.

A natural extension would estimate direction-conditional rewrite diagnostics for each target attribute and rewriter, and report RATE over a plausible range of asymmetry values. Testing several rewriters and prompts would also reveal whether sensitivity is primarily model-specific, instruction-specific, or attribute-specific.

10 Limitations

Our reward function assumes a homogeneous treatment effect, so ATT, ATU, and ATE coincide. The design therefore does not reveal whether direction-dependent rewrite errors interact with treatment-effect heterogeneity.

The off-target attribute is a single binary flag with a fixed linear reward effect. Real LLM rewrites can change continuous and multidimensional properties such as tone, length, and structure, and these changes may interact with the treatment in violation of Assumption 2. Because we enforce Assumption 2, our results isolate Assumption 1 but do not describe joint violations.

Finally, we do not use a real LLM rewriter or reward model. This ensures exact ground truth, controlled asymmetry, reproducibility, and no API dependence, but it prevents us from locating a real system on the Δ axis. Our conclusion concerns RATE's sensitivity to violation of its stated assumption, not the empirical size of that violation in any particular deployed model.

11 Division of Work

Student	Main responsibilities
Tuvia Hausdorff	Paper reading, coding, results analysis, and writing
Yuval Ratzabi	Critique, coding, Git, writing, slides, and video editing

References

Reber, D., Richardson, S. M., Nief, T., Garbacea, C., and Veitch, V. (2025). RATE: Causal Explainability of Reward Models with Imperfect Counterfactuals. *Proceedings of the 42nd International Conference on Machine Learning*, PMLR 267. `https://arxiv.org/abs/2410.13844`.